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عمارة 4 - شارع محمد توفيق دياب - عباس العقاد - مدينة نصر - الدورال 1

(Senior Academy - IT training center)

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DevOps

Engineer Diploma



DevOps



DevOps Engineer Diploma Kubernetes Labs

Kubernetes Scaling, Update Strategies, and Config map and Secret

Lab Objectives

- **Creating and Managing Namespaces**
- **Deploying and Configuring Applications in Kubernetes**
- **Exposing Applications Using Services**
- **Scaling Deployments Manually and Automatically**
- **Configuring and Applying Resource Quotas**
- **Verifying Kubernetes Resources (Pods, Deployments, Services, Quotas)**
- **Managing Environment Variables for Deployments**

Kubernetes Lab: Deploy MySQL Using ConfigMap and Secret

Lab Objectives:

Deploy a MySQL database on Kubernetes using:

- ConfigMap for non-sensitive configuration
- Secret for sensitive data
- Deployment resource for MySQL Pods
- Service resource for internal access

Step 1 - Create Namespace

- Create a namespace named database

Task 2: Create a ConfigMap

Create a ConfigMap named:

mysql-config

Store the following configuration:

Key	Value
MYSQL_DATABASE	companydb
MYSQL_USER	appuser

Task 3: Create a Secret

Create a Secret named:

mysql-secret

Store the following sensitive values:

Key	Value
MYSQL_PASSWORD	StrongPassword123
MYSQL_ROOT_PASSWORD	RootPassword123

-
- **task 4: Create a MySQL Deployment**
 - Create a Deployment named:
 - mysql
 - Requirements:
 - Use the official MySQL image.
 - Deploy 1 replica.
 - Use environment variables from both:
 - ConfigMap
 - Secret
 - Expose container port:
 - 3306
 - Place the Deployment inside the database namespace.
 - Verify that the Pod starts successfully
-

Task 5: Create a Service

Create a Service named:

mysql-service

Requirements:

- Service type: ClusterIP
- Target the MySQL Pod
- Expose port:

3306

Verify service connectivity.

LAB2:

Task 1: Create the Namespace

Create a namespace called:

strategy-lab

Verify its creation.

Task 2: Create the Deployment

Create a Deployment named:

webapp

Requirements:

- Image: nginx
- Replicas: 4
- Deployment strategy: default
- Namespace: strategy-lab

Verify all Pods are running.

Task 3: Perform a Rolling Update

Update the image version.

Task 4: Review Rollout History

Display rollout history for the Deployment.

Identify:

- Current revision
- Previous revision

task 5: Change the Strategy

Modify the Deployment strategy from:

RollingUpdate

to:

Recreate

Verify the change.

Task 6 Perform an Update Using Recreate

Update the image version again.

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